

Hamzeh Hamdan

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Education

Harvard University

Cambridge, MA

B.A., Computer Science & Statistics (Honors). Minor in Mathematics. GPA: 3.72.

May 2025

Thesis: *Cross-Market Signals: Economic Spillovers Across Markets*

- Modeled U.S.-China equity spillovers using linear regressions, sparse additive models, and kernel regressions on 153 JKP factors, explaining up to 80% (U.S.) and 60% (China) market variance.
- Demonstrated asymmetric information flows, showing U.S. indicators significantly improve Chinese return forecasts while Chinese factors add minimal value for U.S. returns.

Activities: Co-led VeritasGPT (a Harvard AI Agent), led a team that sourced \$550k in consulting cases in 4 weeks, researched quantum investment opportunities for NATO and presented at HQ.

Coursework: Probability, Statistical Inference, Linear Models, Algorithms, Systems Programming and Machine Organization, Data Science, Deep Learning for Unstructured Data, SWE with Generative AI.

Experience

Comcast

Philadelphia, PA

Finance Intern

May 2024 – May 2025

- Developed the first internal generative AI tool for finance teams using the Azure OpenAI API.
- Integrated database connections and management, file search, report generation, and data analysis and visualization in the application through a Python user interface.
- Initial testing with equity-based compensation analyses reports increased efficiency by ~93%.

Propel Bio Partners (Biotech Hedge Fund, \$150M AUM)

Remote

Data Science Intern

January – February 2024

- Built a Python class to run Monte Carlo simulations for portfolio-return and risk estimation and created a dashboard for results visualization.
- Provided methods to quantify margin of error on long-term investment success probabilities.

Projects

VeritasGPT

May 2024

- Co-led the development of VeritasGPT, a Harvard AI agent integrating RAG and function calling to access real-time campus data on course catalogs, dining menus, and shuttle timing.
- Engineered prompts and retrieval pipelines to improve context awareness and accuracy.
- Led the Python backend and vector database development, deploying a demo of core features.

Estimating Company Similarity from Embeddings of SEC 10-K Filings

May 2024

- Built a Streamlit web app that uses BERT-style embeddings and cosine similarity to compare company profiles from SEC 10-K subsections, with reusable Python modules for tokenization, clustering, visualization, and data analysis.

Awards

MIT iQuHACK 2024 | 2nd Place in Moody's Challenge

Cambridge, MA

Mean-VaR Portfolio Optimization with Simulated Quantum Annealing

February 2024

- Developed a mean-VaR (value at risk) portfolio optimization algorithm, framed as a quadratic unconstrained binary optimization problem and solved using simulated quantum annealing.

Skills

Technical: Python, Linux, SQL, Azure Services